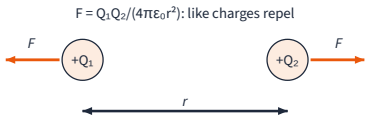
CHAPTER 12 ELECTRO-STATICS

Key idea: charges exert forces through electric fields; potential is the work done per unit charge.

1 COULOMB'S LAW



$$F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r^2}$$

- $\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^9$ N m² C⁻²
- Like charges repel, unlike attract
- In a medium: divide by ε_r
- Several charges: add forces as **vectors**

2 ELECTRIC FIELD

$$E = \frac{F}{q} \qquad E = \frac{Q}{4\pi\varepsilon_0 r^2} \qquad E = \frac{V}{d}$$

- Force per unit **positive** charge; N C⁻¹ = V m⁻¹
- Lines start on +, end on −; never cross

3 GAUSS'S LAW

$$\Phi = \oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\varepsilon_0}$$

Charge distribution	E
Sphere, $r \geq R$	$\frac{Q}{4\pi\varepsilon_0 r^2}$
Infinite line	$\frac{\lambda}{2\pi\varepsilon_0 r}$
Infinite sheet	$\frac{\sigma}{2\varepsilon_0}$
Conductor surface	$\frac{\sigma}{\varepsilon_0}$

Inside a conductor $E = 0$; charge sits on the surface.

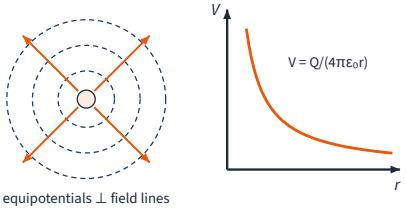
4 ELECTRIC POTENTIAL

$$V = \frac{Q}{4\pi\varepsilon_0 r} \qquad U = qV$$

$$U = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r}$$

- Work done per unit positive charge from **infinity**
- Scalar: add potentials algebraically
- Work by external agent: $W = q\Delta V$

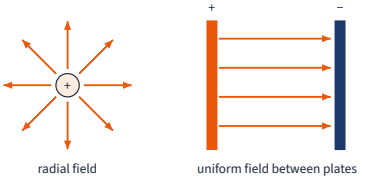
5 FIELD AND POTENTIAL



$$E = -\frac{dV}{dr}$$

- E points towards **lower** potential
- Equipotentials ⊥ field lines; no work along them
- Close equipotentials = strong field

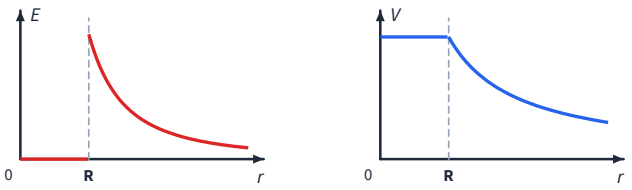
6 CHARGES IN A UNIFORM FIELD



$$a = \frac{qE}{m} \qquad \tfrac{1}{2}mv^2 = qV$$

- Enters ⊥ to the field: **parabolic** path
- Oil drop at rest: $\frac{qV}{d} = mg$

7 CHARGED CONDUCTING SPHERE



Inside: $E = 0$, V constant. Outside: like a point charge at the centre.

8 COMMON SECTION A TRAPS

- Dropping the sign of a negative charge in V (keep it; V is a scalar).
- Adding field magnitudes without directions.
- Using the diameter for r, or µC without ×10⁻⁶.
- $E = 0$ does **not** mean $V = 0$.
- Halving r: F ×4, V ×2.

EXAM TIPS

- ✓Draw each field vector before adding.
- ✓Use energy, $\frac{1}{2}mv^2 = qV$, for speeds.
- ✓State the Gaussian surface you choose.
- ✓Give direction of E as well as magnitude.